

Auction Web App

Cars & Gadgets — Nigerian Market

1. Overview

This is an online auction platform for the Nigerian market, covering two categories: Cars and Gadgets. Registered users — including car dealers, gadget sellers, and individual buyers — can bid on listed items in timed auctions. To keep bidding serious, every bidder must hold a percentage of their bid amount (10–20%) in their dashboard before placing the bid; if they don't have it, the system reminds them to top up ahead of time. Winners have 24 hours to pay the full amount, and payments are processed through OPay.

Uploading (listing an item) is gated by an access code — only approved users see the upload section on their dashboard. This keeps the supply side trusted and verified. Notifications go out via email and WhatsApp when auctions are about to start.

2. User Roles

- **Individual Bidder.** Registers on the platform, maintains a wallet balance, receives notifications, and places bids on cars or gadgets.
- **Car Dealer.** Registered business account that can bid on cars and, if granted an access code, list vehicles for auction.
- **Approved Lister (Car).** Has an access code that unlocks the car-listing section of their dashboard. Works with a verified mechanic to inspect vehicles before listing.
- **Approved Lister (Gadget).** Has an access code that unlocks the gadget-listing section. Must upload proof of ownership for every gadget.
- **Admin.** Issues access codes, reviews listings and proof documents, resolves disputes, and manages platform settings.

3. Access Control

Registration and login use a single flow for all users. The difference is what appears on the dashboard after login:

- Regular users see: browse auctions, wallet, bid history, notification preferences.
- Users with a car access code also see: a “List a Vehicle” section on their dashboard.
- Users with a gadget access code also see: a “List a Gadget” section on their dashboard.

Access codes are issued by admins after the user applies and is approved. A single user can hold both codes.

4. Cars Auction Flow

4.1 Listing a Car

Only users with a car access code can list. Before a car goes live, it must be physically verified. The flow:

- A registered mechanic inspects the vehicle in person.

- The mechanic takes clear photos of the vehicle from multiple angles (exterior, interior, engine, tyres, odometer).
- The lister fills in the car details: make, model, year, colour, registration number, mileage, overall condition, and any known faults (e.g., engine issue, bad AC, body dent).
- All photos and details are uploaded via Openinary.
- The lister sets the auction start time, duration, and the starting (base) price.
- Admin reviews and approves the listing before it goes live.

4.2 Notifications for Car Auctions

When a car auction is scheduled, the platform notifies all registered car dealers and all individual users via email and WhatsApp. The message includes the vehicle summary and the auction start time, with a link to join. The notification goes to everyone in the car audience — there is no opt-out toggle for car auctions in v1 because the audience is already a curated registered base.

4.3 Bidding on a Car

- The auction runs for the duration set by the lister.
- Bids must be at or above the base price.
- Each time a user places a bid, 10–20% of that bid amount is held on their dashboard (the exact percentage is set per-auction by the lister, within the 10–20% range).
- A user cannot place a bid if they don't have enough available balance for the hold. They must top up via OPay first.
- When a user is outbid, their held amount is released back to their available balance.
- When the auction timer ends, the highest bidder wins.

4.4 Payment and Settlement (Cars)

- The winner has 24 hours to pay the full bid amount via OPay.
- The held 10–20% counts toward the final payment; the winner pays the remainder.
- If the winner fails to pay within 24 hours, their held amount is forfeited (per platform policy) and the auction either falls to the second-highest bidder or is re-listed — admin decides based on case.

5. Gadgets Auction Flow

5.1 Listing a Gadget

Only users with a gadget access code can list. Every gadget requires proof of ownership before going live. The flow:

- The lister uploads proof of ownership: the original receipt, OR a police report (if the box/carton or original receipt is missing).
- The lister fills in gadget details: type, brand, model, colour, battery health percentage, specs (RAM, storage, etc.), usage history, and any dents or cracks.
- Photos and short video footage of the gadget are uploaded via Openinary.
- The lister sets the auction start time, duration, and base price.
- Admin reviews the proof of ownership and approves the listing before it goes live.

5.2 Notifications for Gadget Auctions (Opt-In)

This is different from cars. On the gadgets side, users have a “Ready to Bid” toggle in their notification preferences. Only users with this toggle ON receive notifications when a gadget auction is about to start. If the platform has 1,000 users but only 200 have the toggle active, only those 200 are notified — via email and WhatsApp. The message includes the gadget summary, auction start time, and a link to join.

5.3 Bidding on a Gadget

- Auction runs for the duration set by the lister.
- Bids cannot be lower than the base price.
- Users can place bids that are lower or higher than the current top bid, but only the highest bid at close wins — so placing a lower bid is effectively a waste of a hold until someone bids higher.
- Each bid holds 10–20% of the bid amount on the bidder’s dashboard.
- When a user is outbid, their hold is released.
- When the auction ends, the highest bidder wins.

5.4 Payment and Settlement (Gadgets)

- Winner has 24 hours to pay the full amount via OPay.
- Held percentage is applied to the final payment.
- If payment is not completed in 24 hours, the hold is forfeited and the gadget falls to the next-highest bidder or is re-auctioned.
- Once payment clears, the gadget is arranged for delivery to the winner.

6. Wallet and the Hold System

Every user has a dashboard wallet with three visible numbers: total balance, held (locked in active bids), and available (total minus held). Funds move into the wallet through OPay top-ups.

- **Hold on bid.** When a user bids, 10–20% of the bid amount is moved from available to held, atomically, in a single database transaction. If there isn’t enough available balance, the bid is rejected.
- **Release on outbid.** The moment someone else outbids them, the previous bidder’s hold is returned to their available balance automatically.
- **Release on auction cancellation.** If admin cancels an auction, all holds are released.
- **Applied on win.** When a user wins, their hold remains locked and is credited toward the final payment when they pay the balance via OPay.
- **Forfeit on non-payment.** If the winner doesn’t pay within 24 hours, the hold is forfeited per platform policy.

7. Top-Up Reminder Notifications

Because bidding requires funds on hand, the platform sends reminders before an auction starts so users can top up in time. This is the “hours before the bid” notification flow:

- **Timing.** Reminders are scheduled at auction creation. Default intervals: 24 hours before start, 6 hours before start, and 1 hour before start.
- **Audience.** For cars — all registered car dealers and individuals. For gadgets — only users with the “Ready to Bid” toggle on.

- **Who gets the top-up nudge.** Within that audience, users whose available wallet balance is below the expected hold (based on the base price) get an extra message saying: “You may not have enough to bid. Top up via OPay before the auction starts.”
- **Channels.** Email and WhatsApp.

8. Payments

All money movement on the platform goes through OPay. OPay is widely accepted in Nigeria and supports fast transactions, which matters for the 24-hour payment window after winning.

- Top-ups: user initiates an OPay payment that credits their wallet once the webhook confirms settlement.
- Final auction payment: after winning, the user completes payment via OPay within 24 hours. On successful webhook callback, the auction is marked settled and the seller/lister is notified.
- All OPay webhook signatures are verified on the backend to prevent spoofed payment confirmations.
- Every wallet movement is logged as a ledger entry so balances are always auditable.

9. Notifications

The platform sends two types of notifications: email and WhatsApp.

- **Email.** Sent via Resend — auction announcements, top-up reminders, outbid notices, win confirmations, and payment receipts.
- **WhatsApp.** Sent via the WhatsApp Business Cloud API — same event types as email, formatted short and scannable.
- Gadget notifications go only to users with the “Ready to Bid” toggle on.
- Car notifications go to the full registered base (dealers + individuals).
- All notifications are queued and retried on failure.

10. Technology Stack

No alternatives — the stack is locked in:

Layer	Choice
Backend Framework	NestJS (TypeScript)
Runtime	Node.js 22 LTS
HTTP Adapter	Fastify (for higher throughput on live bidding)
Real-time Bidding	Socket.IO over WebSockets, with Redis Pub/Sub adapter for scaling
Database	PostgreSQL
Cache & Job Storage	Redis
Background Jobs	BullMQ (for scheduled notifications, 24-hour payment window, auction closing)

Layer	Choice
Image & File Upload	Openinary
Payments	OPay
Email Notifications	Resend
WhatsApp Notifications	WhatsApp Business Cloud API (via Meta)
Authentication	JWT access + refresh tokens, Argon2 password hashing
Deployment	Docker on Railway (early stage), AWS when scale demands it

Why NestJS: auctions involve concurrent bids, wallet holds, row-level locking, scheduled jobs, and live WebSocket updates. NestJS's modular architecture, dependency injection, and first-class TypeScript support keep this complexity organised. The Fastify adapter gives Nest the raw throughput needed for real-time bidding. Plain Node.js/Express would work, but for a system with this many moving parts, you'd end up rebuilding NestJS yourself.

11. Backend Module Layout

- `auth/` — registration, login, JWT
- `users/` — profiles, access codes, notification preferences (including the Ready-to-Bid toggle)
- `wallets/` — balance, holds, ledger entries
- `cars/` — car listings, mechanic verification fields, photos
- `gadgets/` — gadget listings, proof of ownership, specs, photos
- `auctions/` — auction lifecycle, start/end scheduling, settlement
- `bids/` — bid placement, hold creation, outbid release
- `payments/` — OPay integration, top-ups, webhooks, final settlements
- `uploads/` — Openinary integration for photos and video
- `notifications/` — email (Resend) and WhatsApp Business API
- `jobs/` — BullMQ queues for scheduled reminders, auction closing, 24-hour payment windows
- `gateways/` — Socket.IO gateway for live bid updates
- `admin/` — access code issuance, listing approval, dispute handling

12. Security and Trust

- All traffic over HTTPS.
- Passwords hashed with Argon2.
- OPay webhook signatures verified server-side.
- Access-code-gated listing prevents anonymous users from uploading cars or gadgets.
- Every bid and wallet movement is logged — disputes can be resolved against the record.
- Admin approval on every car and gadget listing adds a human trust layer before items go live.
- Rate limiting on bid submission and payment endpoints.

13. Additional Suggestions

These weren't in the original brief but are worth considering before build:

- **Anti-sniping.** If someone places a bid in the last 2 minutes of an auction, extend the end time by 2 minutes. This prevents users from winning by bidding at the exact last second and keeps the auction fair.
- **Bidder reputation.** Track how often a user wins but fails to pay within 24 hours. Repeat offenders can be suspended or required to hold a higher percentage (e.g., 50% instead of 20%).
- **Verified mechanic directory.** Keep a short list of platform-approved mechanics. Their ID goes on every car listing so buyers know who did the inspection.
- **Live bid feed.** Show bidders a real-time list of bids as they come in (with bidder aliases, not full names) so the auction feels active and transparent.
- **Winner identity check for high-value cars.** For cars above a certain threshold, require a quick BVN/NIN check before payment is accepted — reduces fraud on big-ticket items.

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